

UNIT 4: MOLECULAR GENETICS

4.6 – MEIOSIS & SEXUAL REPRODUCTION

Meiosis

Isn't that just mitosis
twice?

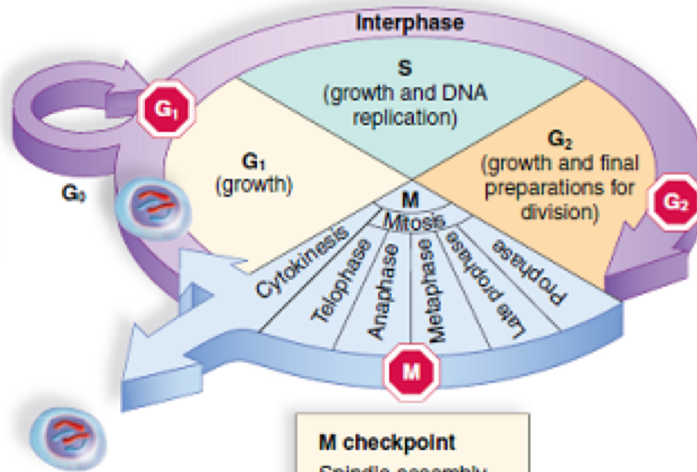
NOPE.



Mitosis Refresher

G₁ checkpoint

Cell cycle main checkpoint. If DNA is damaged, apoptosis will occur. Otherwise, the cell is committed to divide when growth signals are present and nutrients are available.

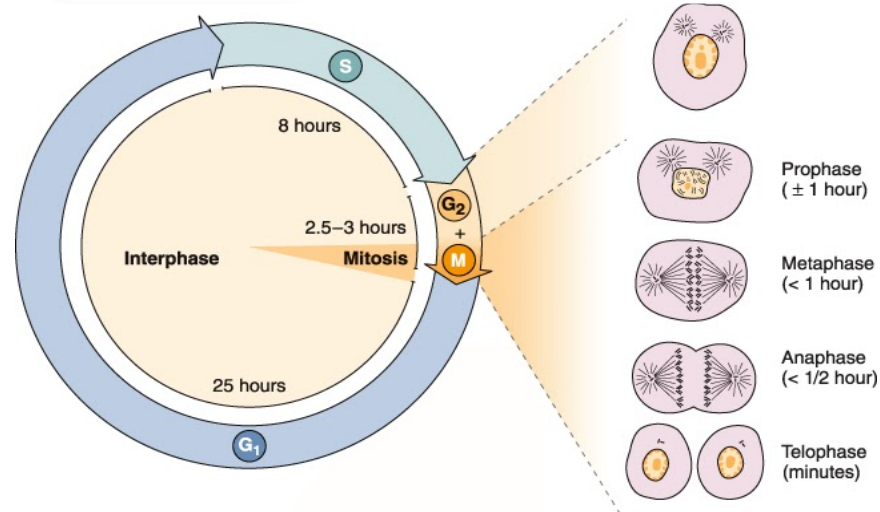
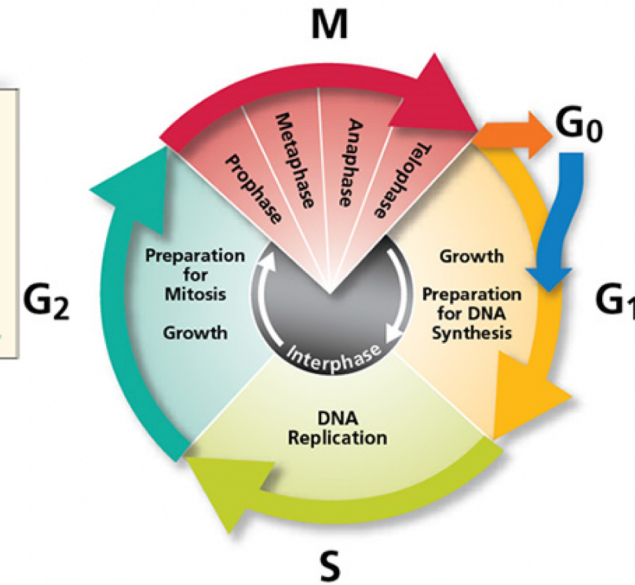


G₂ checkpoint

Mitosis checkpoint. Mitosis will occur if DNA has replicated properly. Apoptosis will occur if the DNA is damaged and cannot be repaired.

M checkpoint

Spindle assembly checkpoint. Mitosis will not continue if chromosomes are not properly aligned.



Intro to Meiosis



- Process of nuclear division that reduces chromosome number in new cells to half original cell
- **Meiosis** produces gametes – haploid reproductive cells
 - ▣ Human **gametes**: sperm and egg
 - ▣ Each contain $1n$ chromosomes (23)
 - ▣ Fusion of sperm & egg produces $2n$ **zygote** with 46 chromosomes
- Cells still undergo G1, S and G2 phases of interphase in preparation of *first division*
- Cells will divide twice:
 - ▣ **Diploid ($2n$) cells produce 4 haploid ($1n$) cells; rather than 2 diploid cells as in mitosis**
- First cell division: **meiosis I**
- Second cell division: **meiosis II**

Meiosis I

- Prophase I:
 - ▣ DNA tightly coils into chromosomes
 - ▣ Spindle fibers appear
 - ▣ Nucleolus & nuclear membrane disassemble/disappear
- Same as in mitosis!

AND THEN



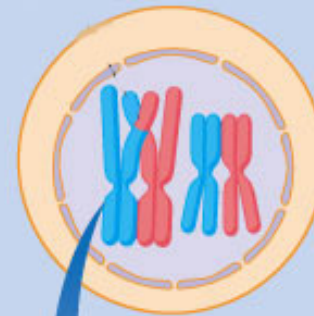
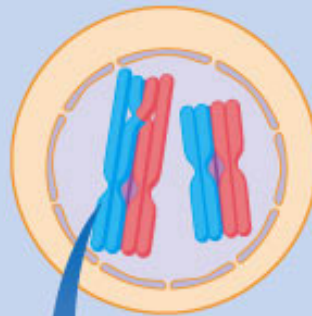
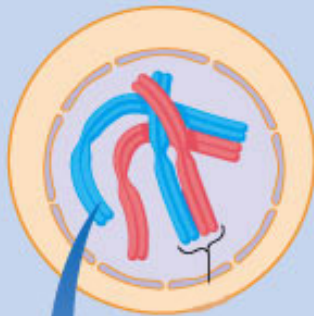
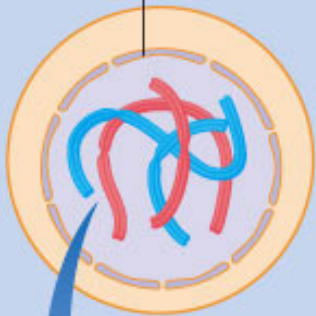
Meiosis I

- Prophase I: What's New?
 - ▣ Chromosomes line up with their homologues
 - **Synapsis:** pairing of homologous chromosomes – forms a tetrad
 - **Tetrad:** pair of homologous chromosomes aligned lengthwise – genes on one chromosome adjacent to same gene on other chromosome
 - ▣ **Crossing-over** occurs during synapsis
 - Chromatids within homologous pair twist around one another
 - Portions may break off; attach to adjacent chromatids on homologous chromosome
 - Allows for exchange of genetic material between maternal/paternal chromosomes
 - Results in **genetic recombination** – new mixture of genetic material
 - Increases amount of genetic diversity

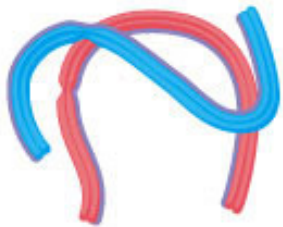
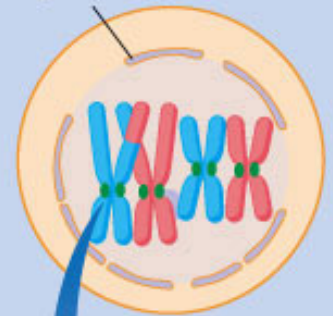
Meiosis I

PROPHASE I

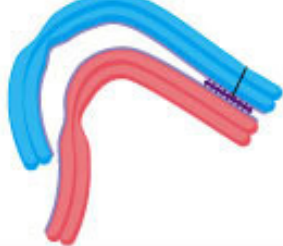
Nuclear envelope



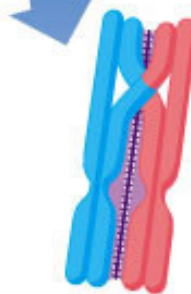
Nuclear envelope fragmenting



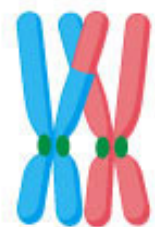
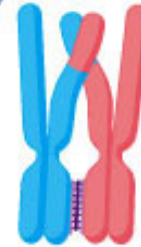
Duplicated chromosomes start to condense



Synapsis begins



Synapsis complete; crossing over occurs



Meiosis I

□ Metaphase I:

- Tetrads line up randomly along midline of dividing cell
- Orientation of chromosome pair random with respect to poles of cell
- Spindle fibers from one pole attach to centromere of one homologous chromosome
- Spindle fibers from opposite pole attach to centromere of other homologous chromosome

□ Anaphase I:

- Each homologous chromosome moves to opposite pole of dividing cell
 - Tetrad is separated; not sister chromatids as in mitosis
- Independent assortment occurs – random separation of homologous chromosomes
 - Adds to genetic diversity/variation

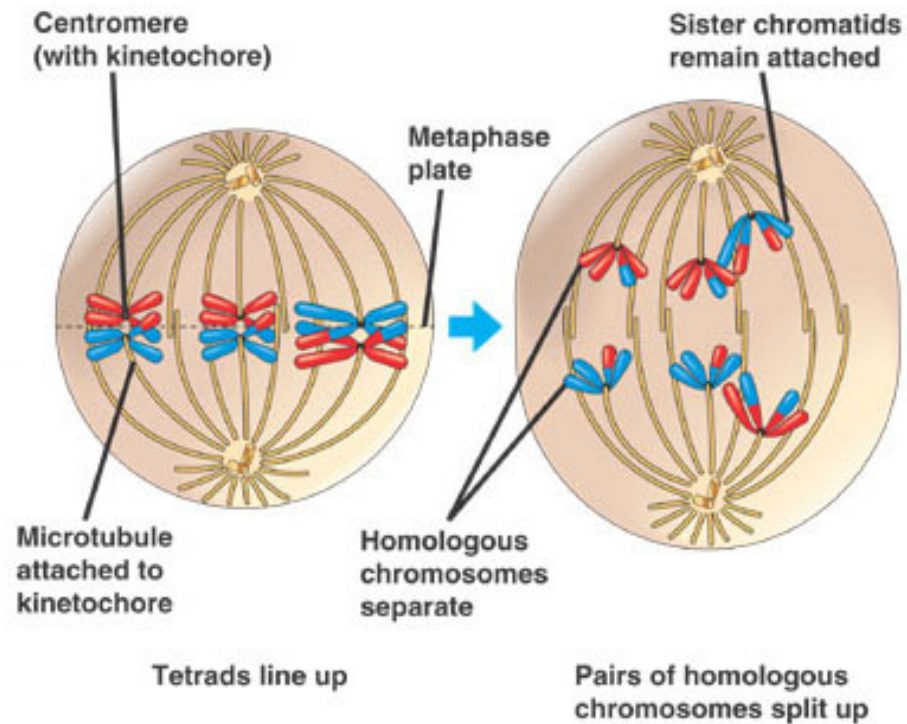


Meiosis I

MEIOSIS I: Separates homologous chromosomes

METAPHASE I

ANAPHASE I



Meiosis I

- Telophase I & Cytokinesis I:
 - ▣ Telophase I – chromosomes reach opposite poles
 - ▣ Cytokinesis begins – cells separate from each other
- Each cell contains 1 chromosome from homologous pair
 - ▣ New cells are $2n - 2$ copies of each chromatid (DNA copied before meiosis began)

IS
THAT IT



In telophase I, nuclear membranes enclose the separated chromatids. The cell membrane completes its constriction.



The first meiotic division ends. There are now two cells, each with the same number of chromatids as the parent cell.



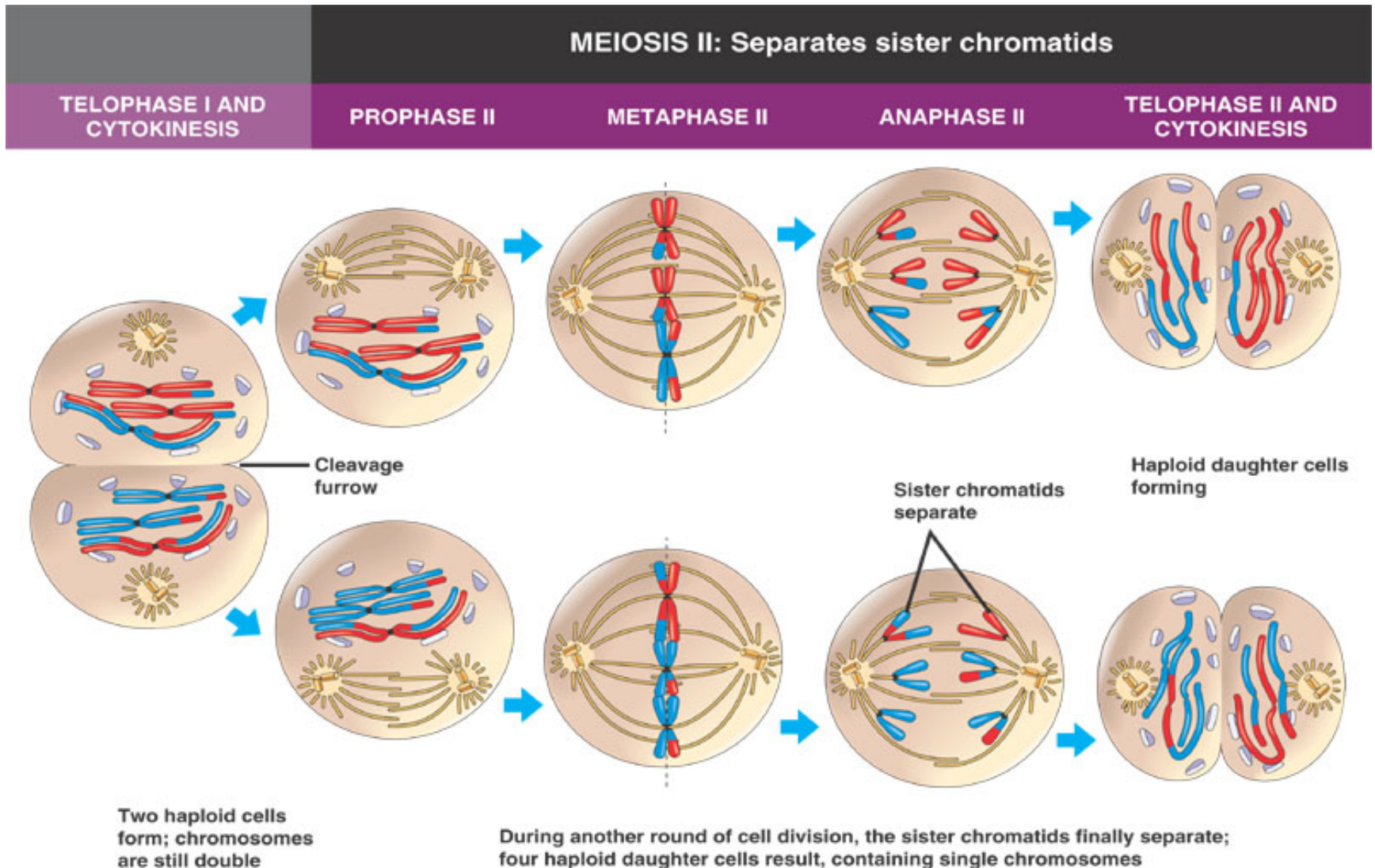
IT'S LIKE MITOSIS!

Meiosis II

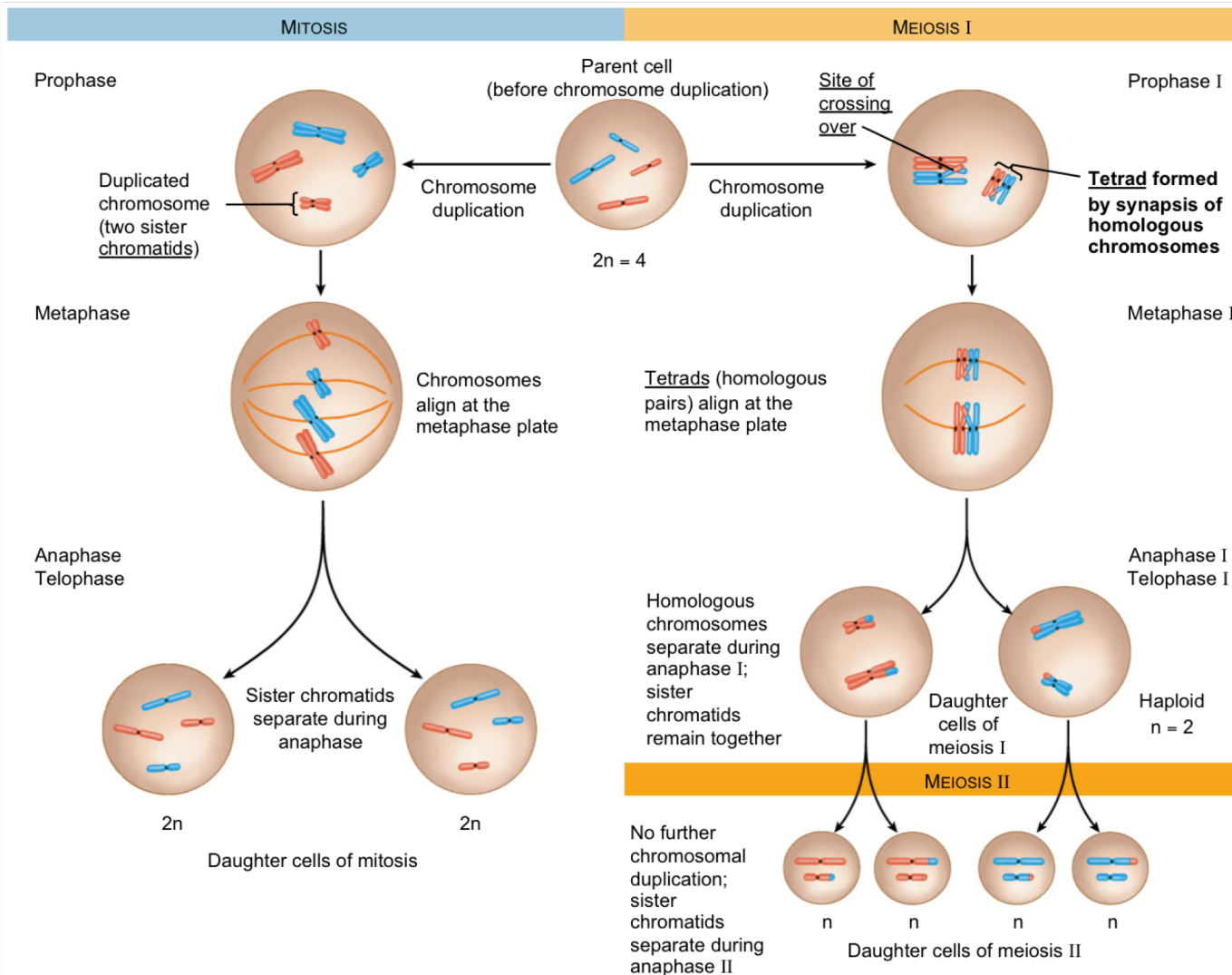


- Occurs in each cell formed during meiosis I
- Not preceded by DNA replication
 - ▣ Cells are still $2n$
 - ▣ Depending on species – can begin after nuclear membrane re-forms or immediately after conclusion of meiosis I
- **Prophase II:** Spindle fibers form, begin to move chromosomes toward midline of dividing cell
- **Metaphase II:** Chromosomes arrive at midline of dividing cell; each chromatid facing opposite poles
- **Anaphase II:** Chromatids separate; move toward opposite poles
- **Telophase II:** Nuclear membrane forms around chromosomes in each of 4 new cells
- **Cytokinesis II:** 4 new cells completely separate; each is haploid
 - ▣ $1n$ – original cell was $2n$

Meiosis II



Mitosis vs Meiosis

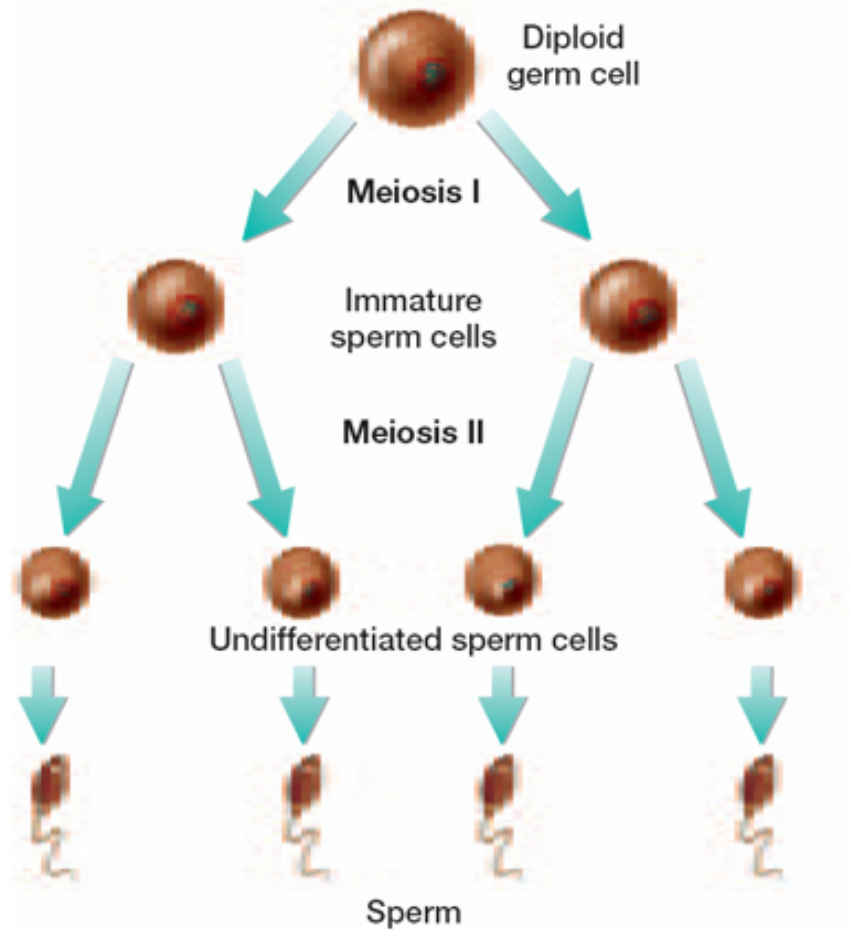


Gametes

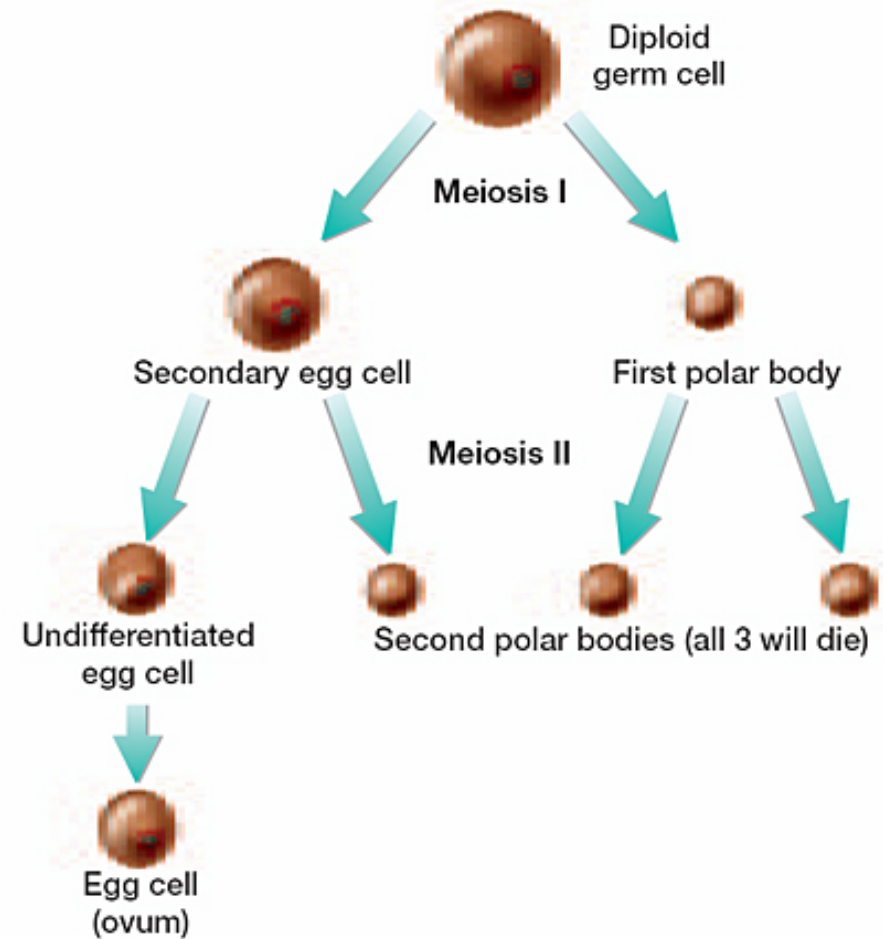
- Animals – only cells that divide by meiosis produce gametes in reproductive organs
 - Timing/structure associated with gamete formation varies
- Human meiosis occurs in:
 - **Testes** (males) to form **spermatozoa** (sperm cells)
 - **Spermatogenesis**: diploid reproductive cell divides meiotically to form 4 haploid spermatids – develops into mature spermatozoa
 - **Ovaries** (females) to form **ova** (egg cells)
 - **Oogenesis**: diploid reproductive cell divides meiotically to produce 1 mature ovum
 - During cytokinesis I & II cytoplasm of original cell divided unequally
 - Ovum receives majority of cytoplasm; other 3 cells produced by meiosis: polar bodies that will degenerate

Gametes

Spermatogenesis



Oogenesis



Male Reproductive System

Organ

Function

Testes

Produce sperm and sex hormones

Epididymides

Sites of maturation and some storage of sperm

Vasa deferentia

Conduct and store sperm

Seminal vesicles

Contribute fluid to semen

Prostate gland

Contributes fluid to semen

Urethra

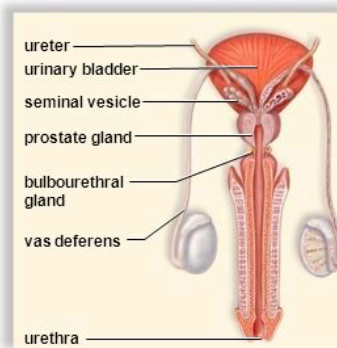
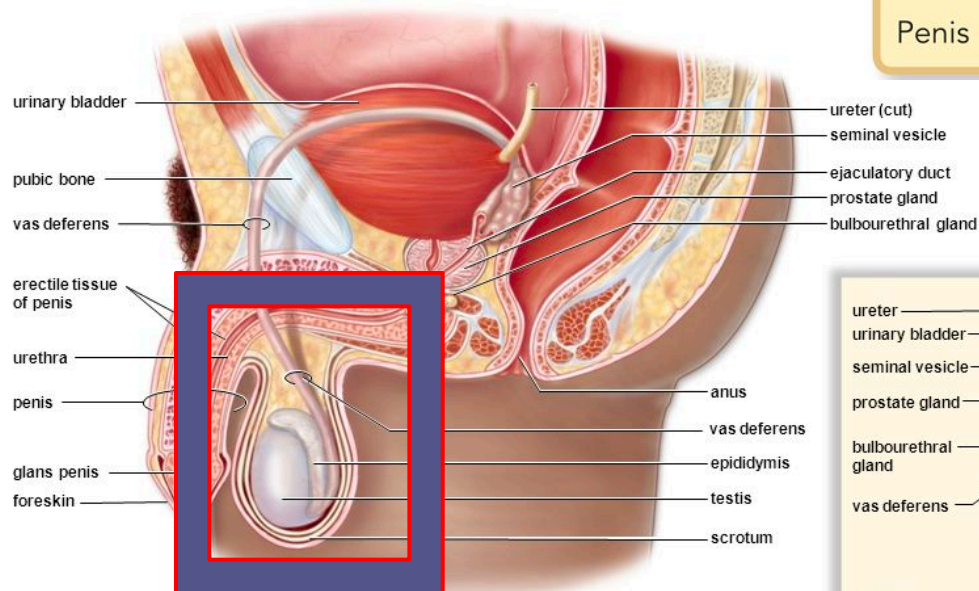
Conducts sperm (and urine)

Bulbourethral glands

Contribute fluid to semen

Penis

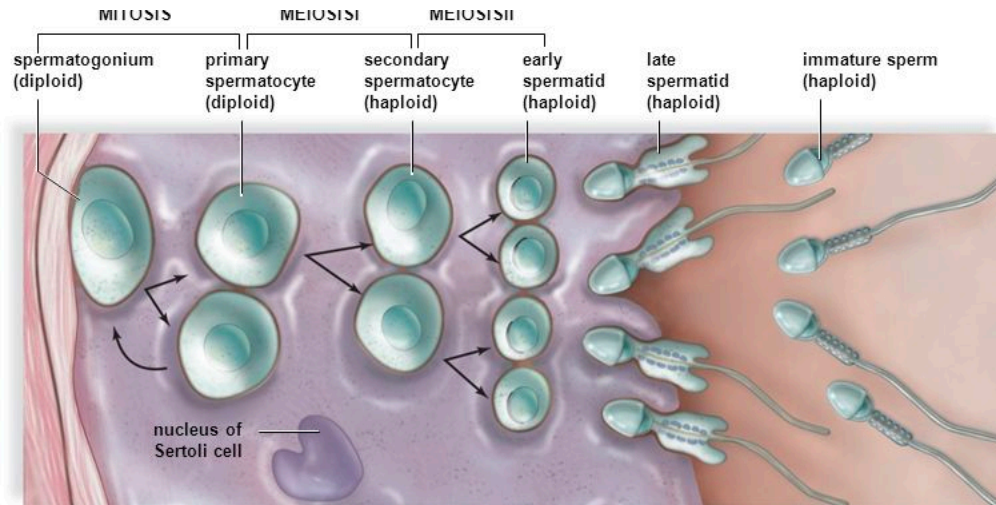
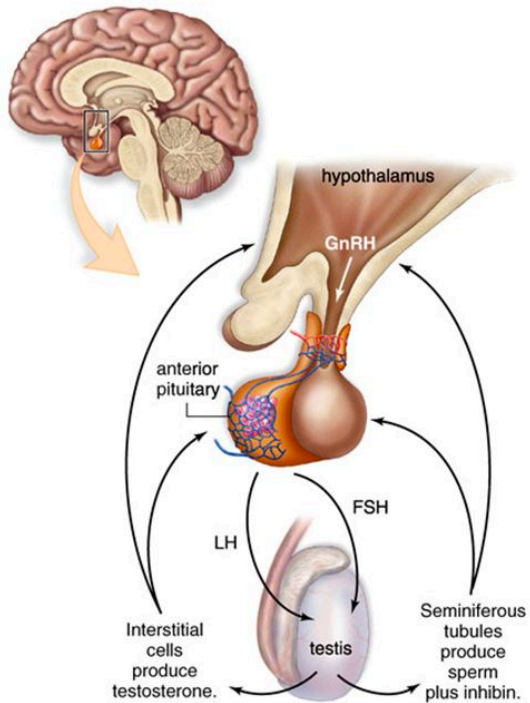
Organ of copulation



Hormones - Males

- Hypothalamus has ultimate control of testicular sexual function
 - ▣ **Gonadotropic-releasing hormone (GnRH)** stimulates production of the gonadotropic hormones by the anterior pituitary
 - ▣ **Follicle Stimulating Hormone (FSH)**
 - Promotes spermatogenesis in the seminiferous tubules
 - ▣ **Luteinizing Hormone (LH)**
 - Controls production of testosterone by the interstitial cells of testes
 - ▣ **Testosterone**
 - Major sex hormone in males
 - Establishes and maintains secondary sex characteristics
 - Responsible for greater muscular development in males

Hormones - Males

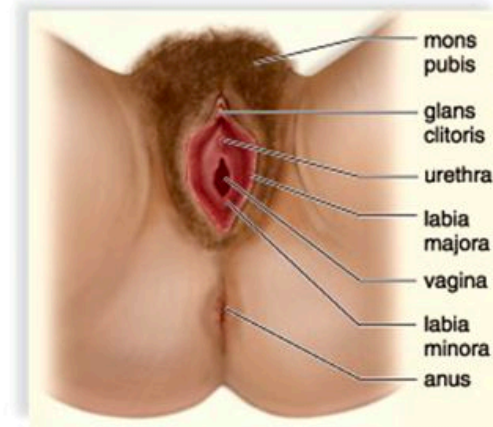
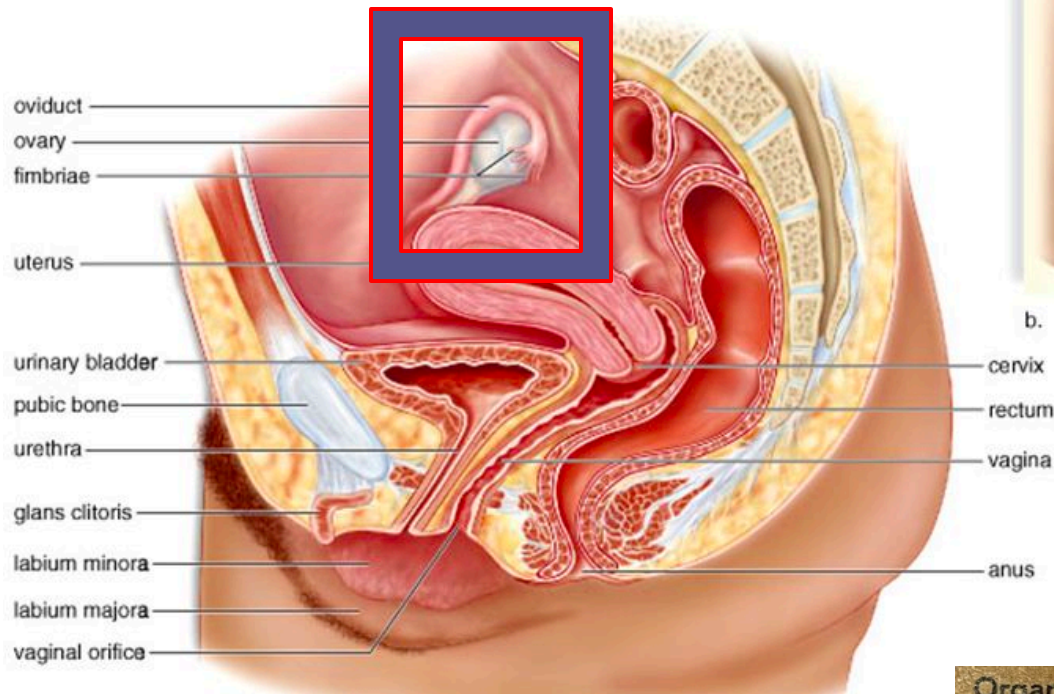


c. Spermatogenesis



d. Sperm

Female Reproductive System



b.

Organ	Function
Ovaries	Produce egg and sex hormones
Oviducts (fallopian tubes)	Conduct egg; location of fertilization
Uterus (womb)	Houses developing embryo and fetus
Vagina	Receives penis during copulation and serves as birth canal

Hormones - Females

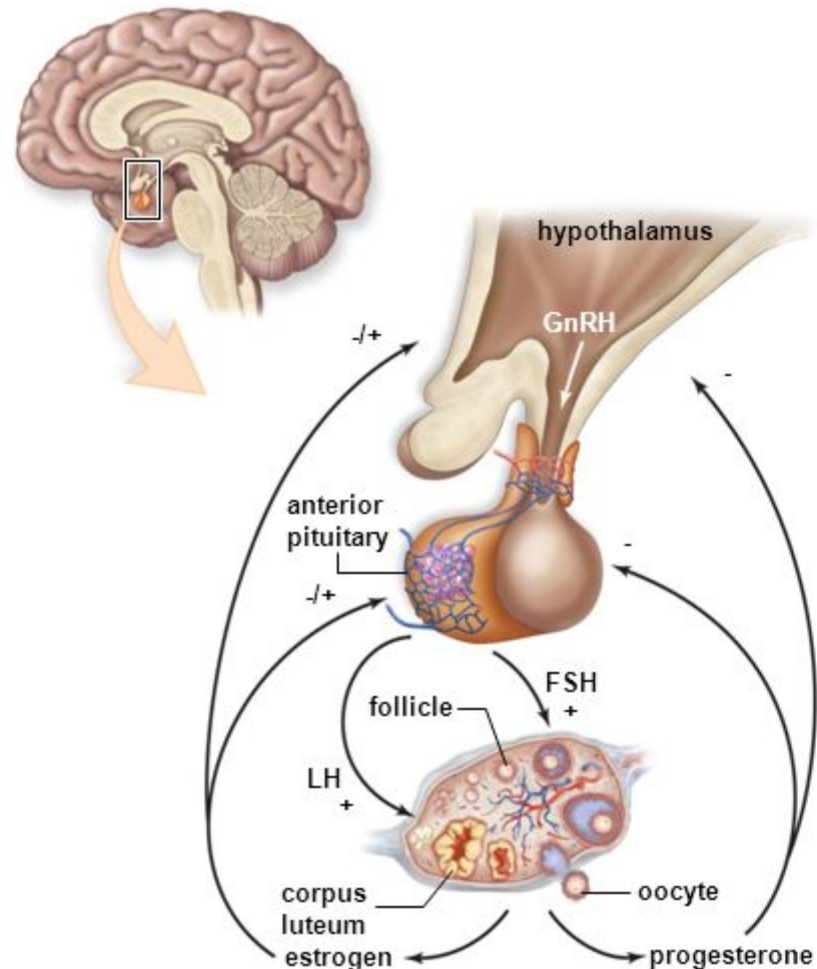
□ The Ovarian Cycle

■ Follicular Phase

- FSH promotes the development of a follicle
- Follicle then secretes estrogens

■ Luteal Phase

- LH promotes development of the corpus luteum
- Corpus luteum then secretes progesterone



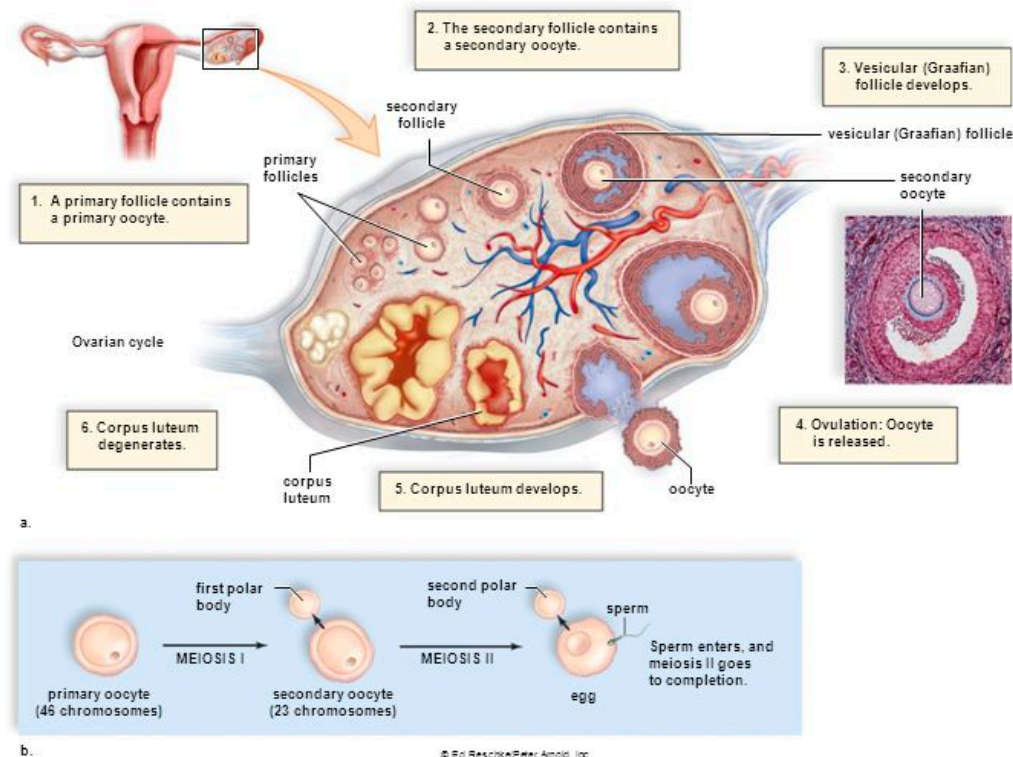
Hormones - Females

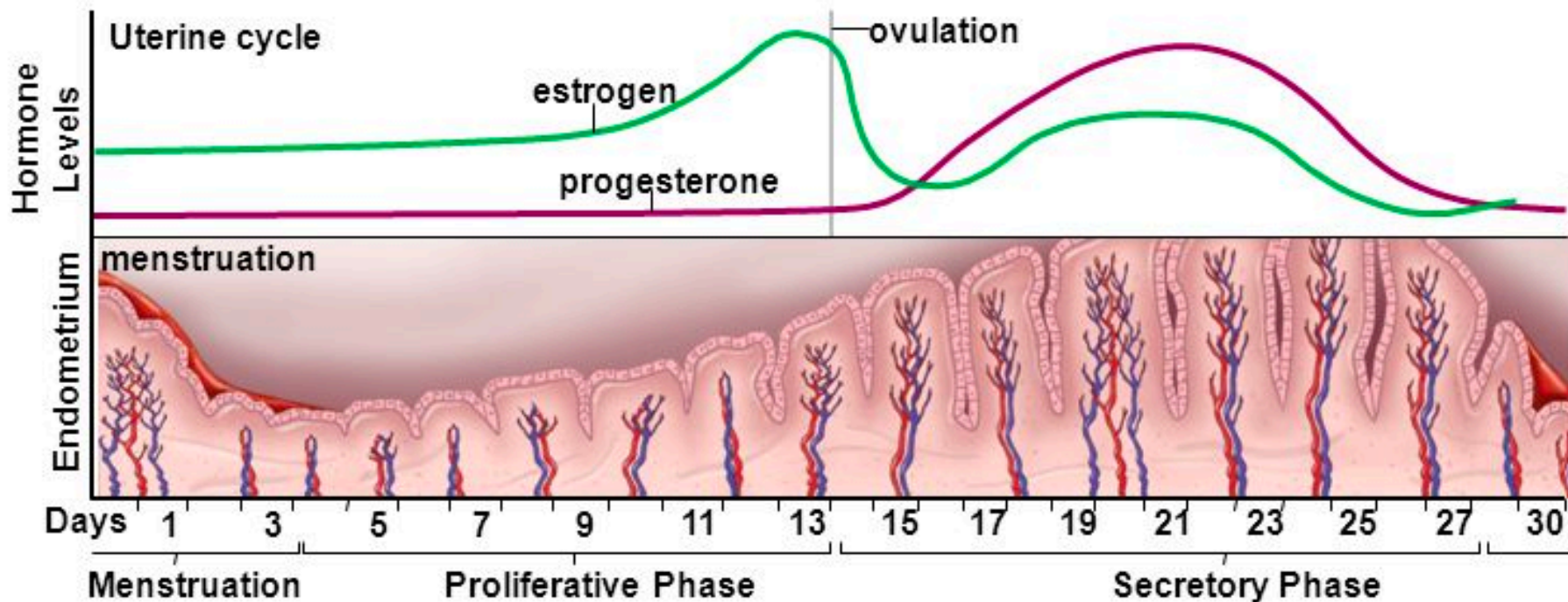
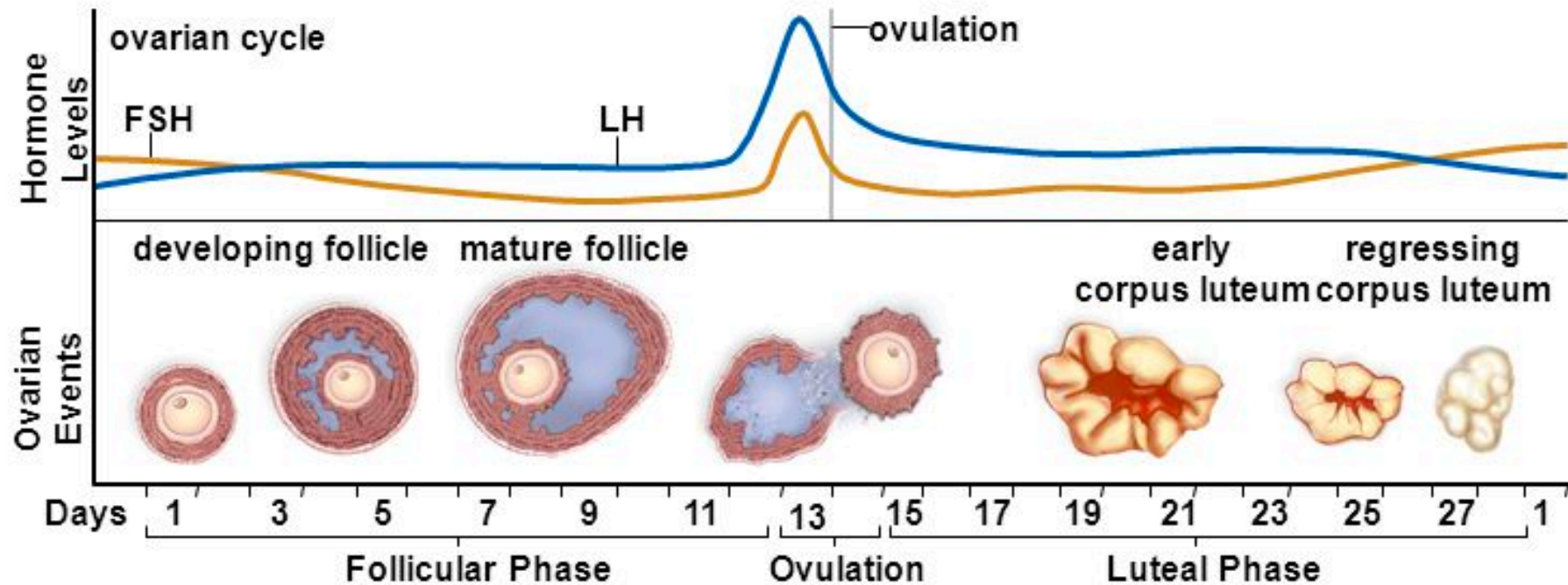
□ The Ovarian Cycle

- Follicle changes from a primary to a secondary to a vesicular (Graafian) follicle under the influence of FSH and LH

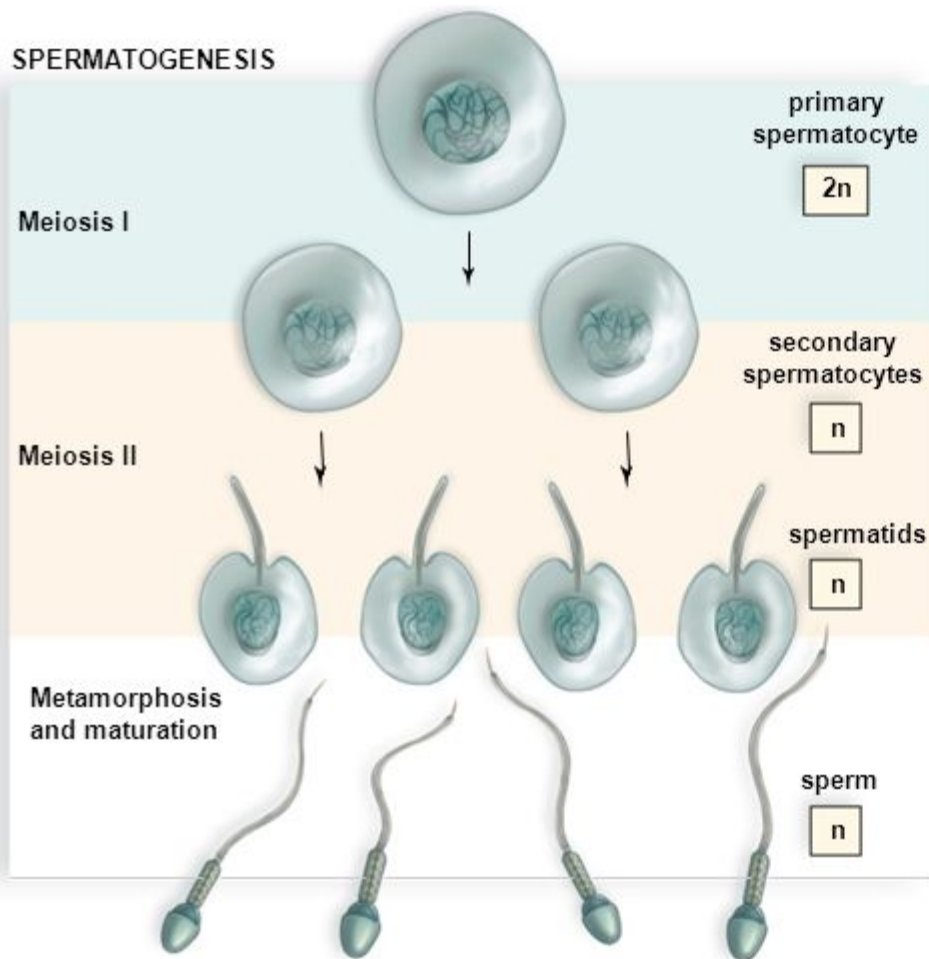
■ Ovulation

- Release of secondary oocyte from a vesicular follicle
- Secondary oocyte enters an oviduct
- Follicle develops in corpus luteum
- Degenerates if pregnancy does not occur
 - If pregnancy does occur, HCG begins to be produced to keep pregnancy viable

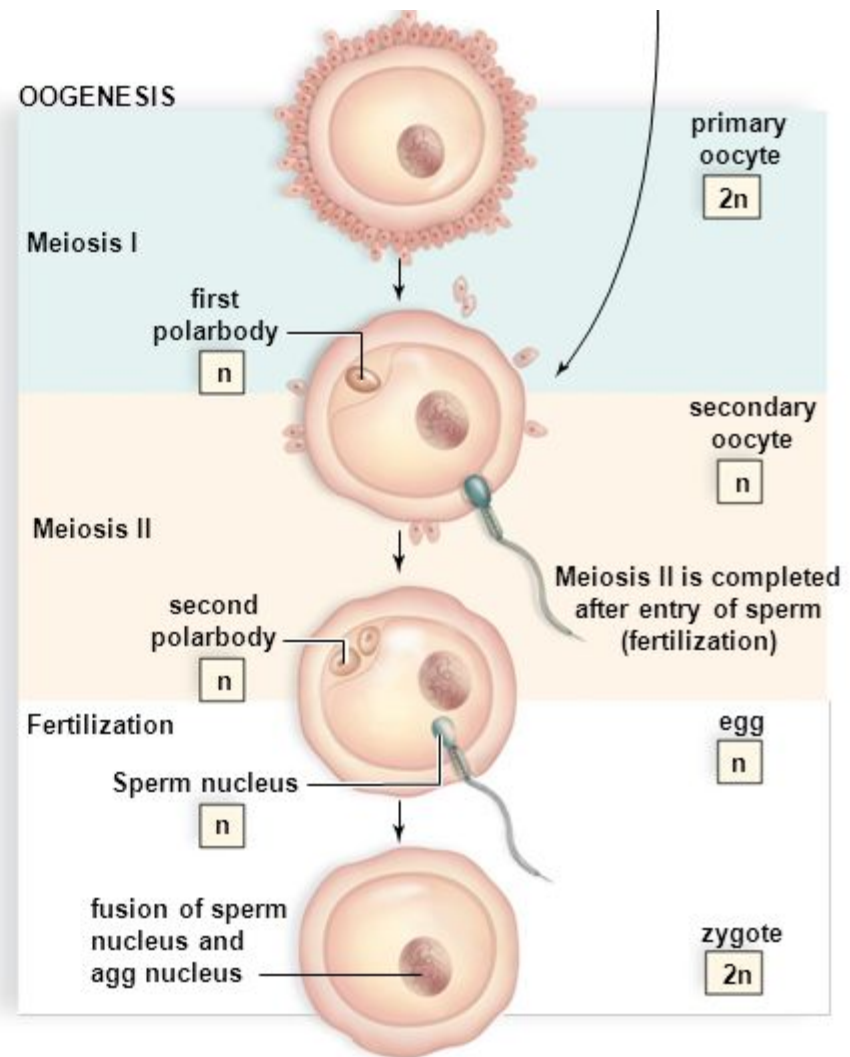




SPERMATOGENESIS



OOGENESIS



Sexual Reproduction

- Production of offspring via meiosis and union of sperm and egg
- Offspring produced are genetically different from parent
 - ▣ Genes are combined in new ways during meiosis
 - Crossing over and independent assortment
 - ▣ Offspring contain unique combinations of parents' genes
 - Except for identical twins
- Evolutionary advantage of sexual reproduction:
 - ▣ Enables species to rapidly adapt to new conditions
 - ▣ Genetic variation leads to 'survival of fittest'
 - Organisms with most favorable traits survive to reproduce and pass on their traits
- Offspring from asexual reproduction are genetically identical to parent (clone)
 - ▣ Offers no genetic diversity
 - ▣ Simplest organisms